Assignment: CNN Development on Custom Dataset

1. Assignment Overview

- Objective: Design and train a CNN to classify images from selected dataset
- Dataset: Select your own dataset with multiple classes
- Submission Format: **CNN_StudetID.ipynb** (GitHub repository link and a copy of the file)
- **Total Marks: 10**
- **Due Date: Apr 02, 2026 (Thursday)**

2. Assignment Requirements

Data Loading and Preprocessing

- Load dataset using PyTorch
- Create training and validation data loaders with batch size

Model Architecture

- Build a custom CNN
- Try with and without batch normalization and dropout for regularization
- Show Document architecture with a detailed summary

Training and Optimization

- Use appropriate optimizer (Adam, SGD)
- Implement learning rate scheduling
- Use cross-entropy loss for multi-class classification
- Plot training and validation loss/accuracy curves

Model Evaluation

- Evaluate on test set and report comprehensive metrics
- Generate confusion matrix visualization
- Analyze per-class performance (best and worst performing classes)
- Save trained model weights

Code Quality & Documentation

- Well-organized, readable code with proper comments
- python notebook with explanations for each section
- Include hyperparameter details and rationale

3. Evaluation Criteria (10 Marks)

Criteria	Description	Marks
Model Architecture	Appropriate CNN design with clear justification	**2**
Data Handling	Correct loading, preprocessing	**1**
Training	Successful training with optimization & regularization	**2**
Test Accuracy	Classification performance on test set	**2**
Evaluation Metrics	Comprehensive metric reporting (Precision, Recall, F1)	**1**
Visualizations	Accuracy and Loss curves, confusion matrix	**1**
Code Quality, Documentation and Saved Model	Well-written, commented code, saved model	**1**

4. Suggested Notebook Structure

1. Import Libraries
2. Load and Explore Dataset
3. Data Preprocessing & Augmentation
4. Define CNN Architecture
5. Training Loop with Validation
6. Evaluate Model on Test Set
7. Visualizations (Loss/Accuracy Curves, Confusion Matrix)
8. Analysis & Discussion of Results
9. Conclusions & Future Work

5. GitHub Should Contain these Deliverables

1. **Python Notebook (.ipynb):** Complete implementation with code, outputs, and visualizations.
   Name: CNN_StudetID.ipynb
2. **Model Weights:** Save trained model in .pth

6. A Microsoft Form will be shared later for the submission of GitHub link and a copy of your ipynb.